Développements limités usuels en 0

Formule de Taylor-Young

À l'ordre n en $a=0$, si $f \in \mathcal{C}^n$:

$$f(x) \underset{x\to 0}{=} f(0) + f'(0)\,x + \frac{f''(0)}{2!}\,x^2 + \cdots + \frac{f^{(n)}(0)}{n!}\,x^n + o(x^n)$$

Puissances

$$(1+x)^\alpha \underset{0}{=} 1 + \alpha x + \frac{\alpha(\alpha-1)}{2}x^2 + \cdots + \frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!}x^n + o(x^n) \quad \alpha \in \mathbb{R}$$

$$\frac{1}{1-x} \underset{0}{=} 1 + x + x^2 + \cdots + x^n + o(x^n)$$

$$\frac{1}{1+x} \underset{0}{=} 1 - x + x^2 - \cdots + (-1)^n x^n + o(x^n)$$

Exponentielle & logarithmes

$$e^x \underset{0}{=} 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!} + o(x^n)$$

$$\ln(1+x) \underset{0}{=} x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots + (-1)^{n+1}\frac{x^n}{n} + o(x^n)$$

$$\ln(1-x) \underset{0}{=} -x - \frac{x^2}{2} - \frac{x^3}{3} - \cdots - \frac{x^n}{n} + o(x^n)$$

Trigonométrie

$$\sin x \underset{0}{=} x - \frac{x^3}{6} + \frac{x^5}{120} - \cdots + (-1)^k\frac{x^{2k+1}}{(2k+1)!} + o(x^{2k+2})$$

$$\cos x \underset{0}{=} 1 - \frac{x^2}{2} + \frac{x^4}{24} - \cdots + (-1)^k\frac{x^{2k}}{(2k)!} + o(x^{2k+1})$$

$$\tan x \underset{0}{=} x + \frac{x^3}{3} + \frac{2x^5}{15} + o(x^5)$$

$$\arctan x \underset{0}{=} x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots + (-1)^k\frac{x^{2k+1}}{2k+1} + o(x^{2k+2})$$

Hyperboliques

$$\operatorname{sh} x \underset{0}{=} x + \frac{x^3}{6} + \frac{x^5}{120} + \cdots + \frac{x^{2k+1}}{(2k+1)!} + o(x^{2k+2})$$

$$\operatorname{ch} x \underset{0}{=} 1 + \frac{x^2}{2} + \frac{x^4}{24} + \cdots + \frac{x^{2k}}{(2k)!} + o(x^{2k+1})$$

Exemple de manipulation

En posant $\alpha = -2$ dans $(1+x)^\alpha$ puis $x \leftarrow -x$:

$$\frac{1}{(1-x)^2} \underset{x\to 0}{=} 1 + 2x + 3x^2 + \cdots + (n+1)x^n + o(x^n)$$